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# AI Assistant for employment rights, contracts, and termination in Slovenia

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**Abstract**

Large language models (LLMs) are powerful but often lack the domain-specific knowledge required for accurate legal advising, particularly in localized contexts such as Slovenian law. In this work, we outline the design of a specialized conversational AI assistant aimed at addressing these limitations. The proposed approach combines the COLESLAW 1.0 legal corpus with techniques such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning (PEFT) to enable context-aware and reliable responses grounded in authoritative sources.

**Keywords**
AI assistant, Slovenian legislation, Retrieval-Augmented Generation

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## Introduction

Although modern large language models are capable of generating fluent and persuasive responses, they are not inherently equipped to handle highly specialized, institution-specific tasks. Their training is based on broad and often globally sourced datasets, which limits their ability to provide precise, localized, and up-to-date information. As a result, they may produce incomplete, inaccurate, or even hallucinated answers when faced with context-dependent queries.

This project addresses these limitations by developing a specialized conversational AI assistant focused on labour law within the Slovenian legal system. The system targets employment-related legal domains such as labour contracts, termination of employment, workers' rights, social security, and related disputes. Rather than relying solely on general-purpose knowledge, the assistant integrates domain-specific legal sources through structured knowledge bases and targeted adaptation techniques, enabling reliable, context-aware legal guidance.

## Related work

In recent years, a growing number of AI-based legal assistants have been developed, particularly in English-speaking countries. One of the most well-known examples is DoNotPay, an AI-powered legal service that automates tasks such as contesting parking tickets and generating legal documents [1]. Another example is PainWorth, a Canadian platform that uses AI to automate personal injury claims [2]. More advanced systems such as Harvey AI employ large language models to assist legal professionals with research, document drafting, and compliance review [3]. These systems illustrate a broad spectrum of AI applications in law, ranging from consumer-facing tools to professional-grade platforms.

In Slovenia, such systems are less common. One notable example is Pravko [4], an AI-based assistant designed to help users navigate legal and administrative procedures. Pravko understands natural language queries and retrieves answers from official sources such as Slovenian legislation and e-government portals. However, existing systems remain limited, highlighting the need for more advanced, data-driven approaches based on large-scale legal datasets and modern language models, which is the focus of our work.

## Methods

By using approaches such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning (PEFT), the assistant will be designed to interpret user intent, retrieve relevant legal information, and generate accurate, context-aware responses. The ultimate goal is to create a reliable AI system that can function as a domain-specific expert, minimizing hallucinations while providing trustworthy and precise guidance.

## COLESLAW 1.0: Database of Slovene legal texts

For the development of our assistant, we use the COLESLAW 1.0 corpus [5], a large-scale collection of Slovenian legal texts. The dataset was published in 2024 as part of the LLM4DH project [6] and is specifically designed to support research in legal natural language processing and domain adaptation of language models.

The corpus contains 547,799 documents with a total of approximately 772 million words. It aggregates documents from multiple authoritative public sources, including the Legal Information System of the Republic of Slovenia (PISRS), court decisions (SodnaPraksa), the Constitutional Court (USRS), and the Official Gazette (Uradni List).

The dataset covers a wide range of legal domains, including enacted and repealed laws, legislative proposals, lower court decisions, regulatory and announcements. Documents are stored in a structured JSONL format and include full cleaned text along with source-specific metadata such as document identifiers, titles, legal basis references, document types and date of decision.

To adapt the corpus for a retrieval-augmented generation (RAG) pipeline and to improve computational efficiency, we further refined the dataset by selecting a relevant subset of documents. We first encoded all documents using a multilingual sentence embedding model, which enables semantic similarity comparison between legal texts and natural language queries.

We then defined a set of 8 representative labour law related queries covering key use cases such as employment contracts, termination procedures, workers' rights, and social security benefits. For each query, we computed cosine similarity scores between the query embedding and all document embeddings in the corpus. Finally, we retrieved the top 750 most relevant documents per query, resulting in a curated subset of the original COLESLAW 1.0 dataset that is better aligned with the target domain and more suitable for efficient RAG-based retrieval.

To guide query design and assess their coverage, we used the thematic classification available in the PISRS system, specifically the section on *Delovno pravo*. This subset was treated as a proxy ground truth for relevant labour law documents. We evaluated the quality of each query by measuring how many of these reference documents were retrieved among its top-ranked results. This approach allowed us to iteratively refine the query set to better capture the target domain and improve retrieval relevance. The final split of relevant and irrelevant documents using this strategy can be seen in Figure 1.

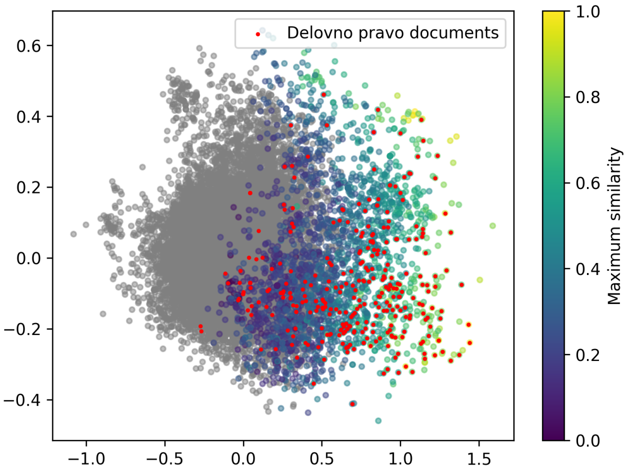


**Figure 1.** Final split of relevant (coloured by maximum query similarity) and irrelevant (grey) PISRS document embeddings in a 2D PCA projection. The proxy ground truth known labour law documents are shown as red dots.

## Retrieval-Augmented Generation

### Document Chunking

To prepare the filtered PISRS documents for retrieval-augmented generation, we apply a chunking strategy tailored to legal text.

Each document is first split according to its natural legal structure. When available, we segment by legal articles (člen), as these represent self-contained legal units. If no article structure is present, the document is treated as a single continuous block.

Large segments are then divided further using paragraph boundaries. Instead of applying a fixed size limit, paragraphs are grouped to form chunks of approximately balanced size. This preserves semantic coherence while avoiding overly fragmented or uneven chunks. To maintain context across neighbouring chunks, chunks are overlapped by one paragraph within segments.

## Language Model and Generation

For the generation component we use Meta Llama-3.1-8B-Instruct [7], an 8 billion parameter instruction-tuned language model. At query time, the top retrieved chunks are formatted into a structured context string and passed to the model alongside a system prompt. The system prompt instructs the model to answer exclusively from the provided legal sources, cite the specific law name and article number, respond in Slovenian, and redirect the user to a lawyer when the answer cannot be found in the retrieved context.

For retrieval we use the `paraphrase-multilingual-mpnet-base-v2` sentence embedding model from the sentence-transformers library (chosen for its multilingual support, including Slovenian). Retrieval uses Maximum Marginal Relevance (MMR) with $k = 5$ chunks and $\lambda = 0.7$, favouring relevance over diversity.

## Results

### RAG Evaluation

To assess the behaviour of the RAG pipeline, we tested the system on a set of employment law questions and manually examined both the generated answers and the retrieved chunks. This allowed us to identify cases where retrieval works well and cases where it fails.

#### Successful Retrieval Cases

The system performed well on questions that map directly to a single, clearly titled law present in the corpus. A representative example is the question *"Katere so obveznosti delodajalca glede varnosti in zdravja pri delu?"*, for which the retriever correctly identified three relevant chunks from the Zakon o varnosti in zdravju pri delu (ZVZD-1). The generated answer accurately summarised the retrieved content, cited the correct law and article, and was written in Slovenian.

#### Problematic Retrieval Cases

Several failures were identified. Questions requiring a specific law not well represented in the corpus produce poor results. For *"Kako dolgo se hranijo evidence o delovnem času?"*, the retriever returned a chunk from a implementing regulation specifically about working hours of road transport workers, rather than a general regulation. The model produced a technically correct but overly narrow answer that does not apply to most workers.

Another problem are questions involving frequently updated legal values such as *"Koliko znaša minimalna plača v Sloveniji?"*. They are problematic because the corpus contains outdated versions of the relevant documents. The retriever correctly identified a document titled *Znesek minimalne plače*, but the retrieved chunk cited the 2015 value of 790.73 EUR, which is no longer current.

### Comparison with MojMaliPravnik

To evaluate the practical usefulness of the system, we compared its answers to legal answers published on Moj Mali Pravnik [8] . For the question *"Koliko dodatnega dopusta mi pripada nad 50 let v kovinski industriji?"*, our system correctly retrieved and cited 26. člen of the Collective Agreement for the Metal Industry of Slovenia including the correct table of additional leave days based on years of working. However, our system incorrectly suggested that being over 50 years old itself entitles the worker to additional leave, while the expert answer correctly clarified that age is not a criterion in this collective agreement.

For *"Si po dopolnjenem 55 letu res starejši delavec in kakšne so tvoje pravice?"*, the answer from Moj Mali Pravnik [8] identified three key rights from ZDR: protection against overtime and night work without written consent, the right to part-time work upon partial retirement, and three additional days of annual leave. Our system retrieved the ZDR, but produced an incomplete answer covering only the part-time work right. It also incorrectly cited a collective agreement provision about healthcare workers over 50 as applying to starejši delavec rights, confusing two different legal concepts.

For *"Delavcu v gostinstvu in turizmu res pripada 1 cel prosti vikend na mesec po zakonu?"*, our system correctly retrieved the relevant Collective Agreement for the Hospitality and Tourism Sector and correctly answered that a full free weekend per month is not guaranteed by law, only 24 hours of uninterrupted weekly rest. The expert answer was more complete, also covering restrictions on irregular working hours for specific groups of workers. Overall, our system produced a concise and accurate answer for this question.

## Conclusion and next steps

We have made significant progress in reducing the source corpus to a subset of documents that are highly relevant to the labour law domain. However, one of the main limitations of the current approach is temporal coverage, as the COLESLAW corpus only includes documents up to 2024. Future work should therefore focus on incorporating more recent legal sources to ensure up-to-date information, as well as implementing mechanisms for validating the continued applicability and legal validity of retrieved documents. An additional challenge lies in distinguishing between regulations that apply broadly to the general population and those that are specific to particular professional or institutional contexts, which will require more fine-grained filtering or classification strategies.

So far, evaluation has been conducted manually on a limited set of models and queries. The next steps include introducing automated evaluation using established metrics such as BLEU and ROUGE, alongside expanding the diversity and complexity of test queries. We also plan to experiment with a wider range of models and configurations in order to systematically compare performance and identify areas where the system requires further improvement.

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